HAREESH LOTTI

Eluru District Nuzvid, AP | 6374248411 | lottiharish@gmail.com | GitHub | LinkedIn | Portfolio

Profile Summary

Emerging **AI/ML Engineer** with hands-on experience building **RAG systems, vector search pipelines**, and **backend APIs** using FastAPI, LangChain, and Qdrant. Skilled in developing scalable document ingestion workflows, semantic search, and intelligent retrieval for real-world AI applications. Strong problem-solver with a focus on clean engineering, automation, and practical AI deployment.

Experience

AI Backend Engineer – Zudu.ai **Nov 2025 – Present**
PRs At Zudu.ai

Product & SaaS-Based Conversational AI Company

- Built and deployed a production-ready **RAG backend microservice** with FastAPI powering a real-time voice assistant.
- Developed a **Hybrid Search system** in Qdrant using dense + sparse scoring, improving technical document retrieval accuracy.
- Improved query relevance with **query rewriting** and optimized threshold logic.
- Designed a robust **document ingestion pipeline** with adaptive text splitting and parent–child indexing for long-context retrieval.
- Added **semantic chunking** and LLM-driven metadata extraction to enhance document understanding.
- Implemented advanced **PDF parsing** with PyMuPDF and OCR fallback for complex, image-heavy documents.
- Built a **Background Sound Mixer** for LiveKit agents, enabling real-time ambient audio overlay across 9 sound types with adjustable volume.

Summer Internship - Black Bucks – AI & ML Intern **2025**

- Developed and optimized machine learning models for real-world business use cases, contributing to core product features.
- Engineered data preprocessing and feature engineering pipelines to enhance model accuracy and robustness.

Web Team Member – eCrush, IIIT Nuzvid **2022 – 2024**

- Developed web automation tools, including a results scraper, to reduce manual work and support student engagement.

Project Contributor – IIIT National Level Fest **2021 – 2025**

- Designed a Lottery Ticket & Email Automation System, generating over **₹60,000 revenue** and adopted annually by juniors.

Projects

Ask Chatbot! – RAG & Multi-Server Agentic AI System *{Generative AI, RAG, MCP, LangGraph, Gemini, Manim, AsyncIO, SQL, LangSmith}*
RAG Repo **Agent Repo**

- Built a high-performance **RAG system** with semantic search and adaptive chunking using LangChain and FAISS for accurate multi-document retrieval.
- Orchestrated **Model Context Protocol (MCP)** to integrate multi-server tools (Math, Expenses, Manim), enabling autonomous animation generation and complex tool execution.
- Engineered a custom **async event-loop backend** to bridge LangGraph's async execution with synchronous Streamlit UI, supporting real-time **streaming and tool status tracking**.
- Implemented persistent session management using **aiosqlite** and ensured observability through **LangSmith** tracing and performance analysis.

Pneumonia Detection from Chest X-Ray Images using Custom CNN Architecture *{Deep Learning,*

Medical Imaging} GitHub | Project Paper

- Developed a 5-layer **Custom CNN architecture** in TensorFlow/Keras for automated pneumonia detection from pediatric chest X-rays.
- Achieved **90.4% test accuracy**, with **93% precision for Pneumonia** and **87% for Normal** cases on the test dataset of 5,863 images.
- Implemented **comprehensive data augmentation** and **Batch Normalization/Dropout** techniques to counter class imbalance and overfitting.
- Proposed future enhancements including **ensemble methods, interpretability with Grad-CAM, and clinical validation**.

StudyBud — Collaborative Study Platform *{Django 5.2, PostgreSQL, Supabase, Docker, Render, HTMX, GitHub Actions}*
GitHub | **Live Demo**

- Developed a full-featured study platform with **real-time room-based messaging** powered by **HTMX** and optimized database queries to eliminate N+1 latency.
- Automated the CI/CD pipeline using **GitHub Actions** for seamless **Docker** image builds and zero-downtime deployments to **Render**.
- Integrated **Supabase** for managed PostgreSQL and S3-compatible media storage, engineering dynamic environment routing for instant local SQLite development vs. production cloud routing.
- Implemented automated user profiling systems using **Django Signals** and engineered a custom, fully-responsive UI with **Vanilla CSS**.

Insurance Premium Prediction API *{ML, FastAPI, Docker}* **GitHub** | **Live API Endpoint**

- Engineered features (BMI, age group, lifestyle risk) and trained a **Random Forest** model to predict insurance categories, achieving **90% accuracy**.
- Built a production-ready REST API using **FastAPI** to serve the trained model and provide real-time inference.
- Implemented server-side data validation and automatic feature computation from raw inputs (e.g., age, height) using **Pydantic** models and `@computed_field` decorators.
- Designed a clear API response schema to return the predicted category, confidence score, and class probabilities.
- Containerized the complete application, model, and dependencies using **Docker** for scalable and isolated deployment.

Education

Bachelor of Technology in Computer Science and Engineering **2022–2026**
Rajiv Gandhi University of Knowledge Technologies (IIIT Nuzvid), Andhra Pradesh *CGPA: 8.85*

Pre-University Course – PUC **2020–2022**
Rajiv Gandhi University of Knowledge Technologies (IIIT Nuzvid), Andhra Pradesh *CGPA: 9.64*

Board of Secondary Education, Andhra Pradesh **2019–2020**
Vignan Vidhya Nikethan *CGPA: 10*

Technical Skills

Programming: Python (advanced), SQL

AI/ML Frameworks: PyTorch, TensorFlow, Scikit-learn, Keras, Transformers

Generative AI: LangChain, LangGraph, LangSmith, **Model Context Protocol (MCP)**, RAG Agents, OpenAI/Gemini/Groq APIs

NLP: Hugging Face, NLTK, spaCy, fuzzywuzzy, BeautifulSoup, SentenceTransformers

Data Analysis & Visualization: Manim (Animation), Pandas, NumPy, SciPy, Matplotlib, Seaborn, Plotly

Web Development & APIs: Django, Streamlit, Flask, FastAPI, **HTMX**, **Multi-server MCP Orchestration**, aiosqlite, REST APIs, HTML/CSS/JS

MLOps & Deployment: Docker, Git, **GitHub Actions**, Hugging Face Hub
Databases: PostgreSQL, **Supabase**, MySQL, Vector Databases (Qdrant, FAISS, Chroma, Pinecone)

Certifications & Achievements

Certifications

- NPTEL – Deep Learning
- NPTEL – Large Language Models (LLMs)
- View All Certificates

Achievements

- 1st Place – Madcode Coding Competition
- 1st Place – Tech Hunt Competition
- Organized and mentored AI/ML workshops for juniors
- View Achievement Proofs